

Titanic Dataset - EDA Summary Report

1. Survival Overview

- Around **38%** of the passengers survived.
- The dataset is **imbalanced**, with more non-survivors than survivors.

2. Gender and Survival

- **Females had a significantly higher survival rate** than males.
- The majority of **male passengers did not survive**, highlighting gender-based evacuation priority (“women and children first”).

3. Passenger Class (Pclass)

- **Survival rates were highest in 1st class**, followed by 2nd class, and lowest in 3rd class.
- Passengers in higher classes likely had **better access to lifeboats** and crew.

4. Age Distribution

- Age ranged from **0.42 to 80 years**.
- **Younger passengers** had slightly better survival chances.
- **Infants and children** had particularly high survival rates, consistent with emergency protocols.

5. Fare Analysis

- Passengers who paid **higher fares** were more likely to survive.
- This suggests a relationship between **fare → class → survival**.

6. Embarkation Port

- Most passengers boarded at **Southampton ('S')**.
- Passengers from **Cherbourg ('C')** had **higher survival rates**, likely due to more 1st class passengers boarding there.

7. Correlation Insights

- Strongest positive correlation with survival:
 - **Fare** (higher fare, more survival)
 - **Being female**
 - **Lower Pclass value** (1st class)
- **Negative correlation** between Pclass and survival (higher class number, lower survival).

8. Missing Data

- **Age** and **Cabin** columns have missing data.
- **Cabin** is missing in ~77% of records, and may need to be **excluded or imputed** cautiously.
- **Embarked** has a few missing values and can be filled with the most common port.